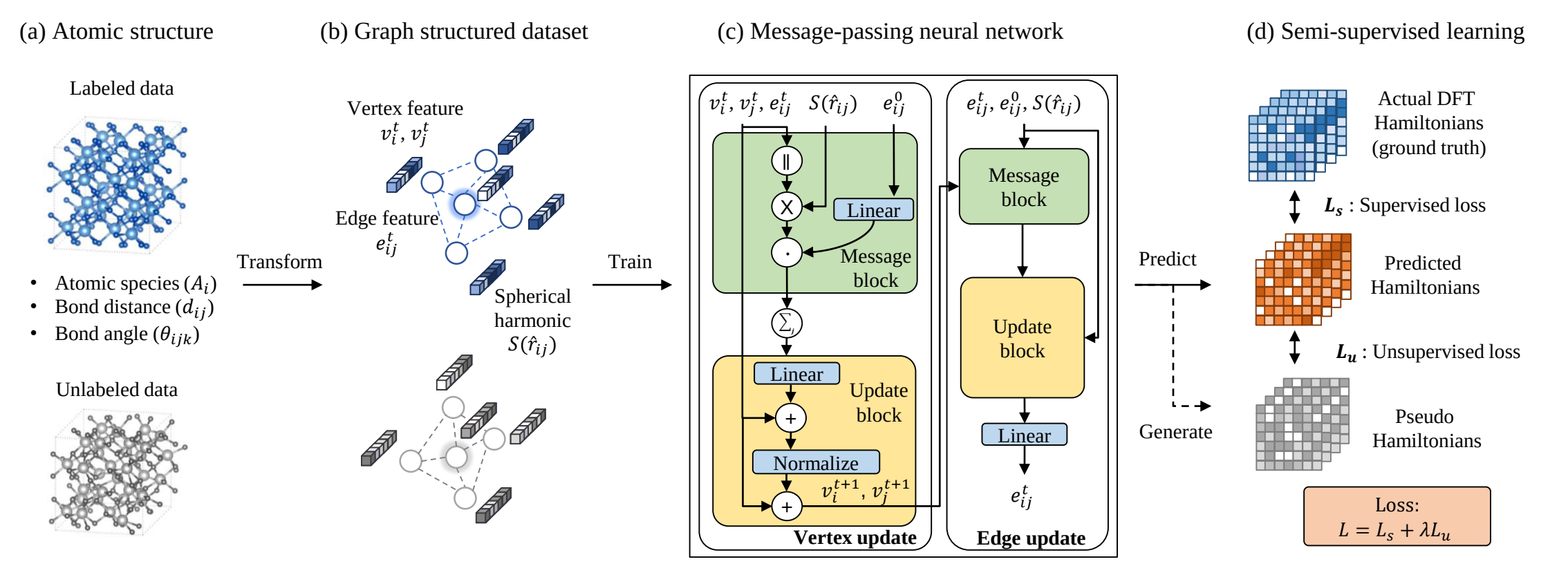

(a) Atomic structure
(b) Graph structured dataset
(c) Message-passing neural network
(d) Semi-supervised learning
Labeled data
Atomic species (A_i)
Bond distance (d_{ij})
Bond angle (θ_{ijk})
Unlabeled data
Transform
Vertex feature
v_i^t, v_j^t
Edge feature
e_{ij}^t
Spherical harmonic
$S(\hat{r}_{ij})$
Train
v_i^t, v_j^t, e_{ij}^t $S(\hat{r}_{ij})$ e_{ij}^0
Linear
Message block
Linear
Update block
Normalize
v_i^{t+1}, v_j^{t+1}
Vertex update
$e_{ij}^t, e_{ij}^0, S(\hat{r}_{ij})$
Message block
Update block
Linear
e_{ij}^t
Edge update
Predict
Generate
Actual DFT Hamiltonians (ground truth)
L_s : Supervised loss
Predicted Hamiltonians
L_u : Unsupervised loss
Pseudo Hamiltonians
Loss:
$L = L_s + \lambda L_u$